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Lab Values, Normal Adult

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Laboratory Reference Ranges in Healthy Adults

The values listed below are generalizations. Each laboratory has specific reference ranges. [1, 2, 3, 4, 8, 9]

Electrolytes

Normal ranges for electrolytes include the following [1, 8, 9] :

- Ammonia: 10-80 mcg/dL or 6-47 µmol/L
- Ceruloplasmin: 23-50 mg/dL or 230-500 mg/L
- Chloride: 98-106 mEq/L or 98-106 mmol/L
- Copper: 100-200 µg/dL, in urine: 0-100 µg/24 h
- Creatinine: 0.8-1.3 mg/dL; males: 0.6-1.2 mg/dL or 53-106 µmol/L; females: 0.5-1.1 mg/dL or 44-97 µmol/L
- Blood urea nitrogen: 10-20 mg/dL or 3.6-7.1 mmol/L
- Ferritin: Females: 10-150 ng/mL or 10-150 mcg/L; males: 12-300 ng/mL or 12-300 mcg/L
- Glucose: 74-106 mg/dL
- Inorganic phosphorous: 3-4.5 mg/dL or 0.97-1.45 mmol/L (SI units)
- Ionized calcium: 4.5-5.6 mg/dL or 1.05-1.3 mmol/L
- Magnesium: 1.3-2.1 mEq/L or 0.65-1.05 mmol/L
- Phosphate: 3-4.5 mg/dL or 0.97-1.45 mmol/L
- Potassium: 3.5-5.0 mEq/L or 3.5-5 mmol/L
- Pyruvate: 0.08-0.16 mmol/L [9]
- Sodium: 136-145 mEq/L or 136-145 mmol/L
- Total calcium: 2-2.6 mmol/L (9-10.5 mg/dL) or 2.25-2.62 mmol/L
- Total iron-binding capacity: 250-310 µg/dL

- Total serum iron: 50-150 µg/dL
- Transferrin: 215-365 mg/dL or 2.15-3.65 g/L
- Urea: 1.2-3 mmol/L
- Uric acid: Blood: Males: 4-8.5 mg/dL or 0.24-0.51 mmol/L; females: 2.7-7.3 mg/dL or 0.16-0.43 mmol/L. Urine: 250-750 mg/24 h or 1.48-4.43 mmol/day
- Zinc: 75-140 µg/dL

Hematology

Normal ranges for hematology values include the following [4, 5, 6, 7] :

- Hemoglobin: Females: 12-16 g/dL or 7.4-9.9 mmol/L; males: 14-18 g/dL or 8.7-11.2 mmol/L
- Hematocrit: Females: 37-47%, volume fraction: 0.37-0.47; males: 42-50%, volume fraction: 0.42-0.52
- Glycosylated hemoglobin: 4.0-5.9%
- Mean corpuscular volume: 80-95 fL
- Red blood cell distribution width: 11-14.5%
- Mean corpuscular hemoglobin: 27-31 pg
- Mean corpuscular hemoglobin concentration: 32-36 g/dL or 32-36%
- Reticulocytes:0.5-2% of red cells
- White blood cells: 5000-10,000/mm^3 or 5-10 × 10^9/L
- Neutrophils: 2500-8000/mm^3 or 55-70%
- Bands: < 1 × 10^9/L
- Lymphocytes: 1000-4000/mm^3 or 20-40%
- Monocytes: 100-700/mm^3 or 2-8%
- Eosinophils: 50-500/mm^3 or 1-4%
- Platelets: 150,000-400,000/mm^3 (150-400 × 10^9/L)
- Prothrombin time: 11-12.5 s
- International normalized ratio: 0.8-1.1
- Activated partial thromboplastin time: 30-40 s
- Fibrinogen: 200-400 mg/dL or 2-4 g/L
- Bleeding time: < 9 min

Lipids

Normal ranges for lipids include the following [8] :

- Triglycerides: Males: 40-160 mg/dL or 0.45-1.81 mmol/L; females: 35-135 mg/dL or 0.40-1.52 mmol/L
- Total cholesterol: < 200 mg/dL or < 5.2 mmol/L
- High-density lipoprotein: Females: > 55 mg/dL or > 0.91 mmol/L; males: > 45 mg/dL or > 0.75 mmol/L
- Low-density lipoprotein: < 130 mg/dL

Acid base

Normal acid-base ranges include the following [8] :

- pH: 7.35-7.45
- Base excess: 0 ± 2 mEq/L
- H+: 36-44 nmol/L
- Partial pressure of oxygen (pO2): 80-100 mm Hg
- Oxygen saturation: ≥ 95%
- Partial pressure of carbon dioxide (pCO2): 35-45 mm Hg
- Bicarbonate(HCO3): 21-28 mEq/L

Gastrointestinal function

Normal gastrointestinal function ranges include the following [8] :

- Albumin: 3.5-5.5 g/dL
- Alkaline phosphatase: 30-120 U/L
- Alanine aminotransferase: 4-36 U/L
- Amylase: 60-120 Somogyi units/dL or 30-220 units/L
- Aspartate aminotransferase: 0-35 U/L
- Direct bilirubin: 0.1-0.3 mg/dL or 1.7-5.1 µmol/L
- γ glutamyl transferase: Males and females aged 45 years and older: 8-38 units/L or 8-38 international units/L. Females younger than 45 years: 5-27 units/L or 5-27 international units/L
- Lipase: 0-160 units/L
- Total bilirubin: 0.3-1.0 mg/dL or 5.1-17 µmol/L
- Total protein: 6.4-8.3 g/dL

Cardiac enzymes

Normal cardiac enzyme ranges include the following [8] :

- Creatine kinase: 20-200 U/L
- Creatine kinase MB (CKMB): < 5% of total CK levels
- Cardiac Troponin: Cardiac troponin T: < 0.1 ng/mL
  - Cardiac troponin I: < 0.03 ng/mL
  - hsTnT: Males: < 22 ng/L; females: < 14ng/L

Hormones

Normal hormone ranges include the following [8, 9] :

- 17-hydroxyprogesterone (female, follicular): < 80 ng/dL
- Adrenocorticotropic hormone: Males aged 19 years and older: 7-69 pg/mL; females aged 19 years and older: 6-58 pg/mL
- Estradiol: Adult males: 10-50 pg/mL; adult females, follicular phase: 20-350 pg/mL, midcycle peak: 150-750 pg/mL, luteal phase: 30-450 pg/mL, postmenopause: ≤ 20 pg/mL
- Free T3: 2.3-4.2 pg/mL [9]
- Free T4: 0.8-1.8 ng/dL [9]
- Follicle-stimulating hormone: Males: 1.42-15.4 IU/L; females, follicular phase: 1.37-9.9 IU/L, ovulatory peak: 6.17-17.2 IU/L, luteal phase: 1.09-9.2 IU/L, postmenopause: 19.3-100.6 IU/L
- Growth hormone (fasting): Males: < 5 ng/mL (mcg/L [SI units]); females: < 10 ng/mL (mcg/L [SI units])
- Progesterone: Adult males: 10-50 ng/dL; adult females, follicular phase: < 50 ng/dL, luteal: 300-2500 ng/dL, postmenopausal: < 40 ng/dL. In pregnancy, first trimester: 725-4400 ng/dL, second trimester: 1950-8250 ng/dL, third trimester: 6500-22,900 ng/dL
- Prolactin: Adult males: 3-13 ng/mL; adult females: 3-27 ng/mL; pregnant females: 20-400 ng/mL
- Testosterone: Males, 291-1100 ng/dL; females: 18-54 ng/dL
- Thyroxine-binding globulin: 12-27 µg/mL
- Thyroid-stimulating hormone: 0.5-4.0 µU/mL (0.5-4.0 mU/L)
- Total T4: Adult males: 4-12 mcg/dL or 59-135 nmol/L; adult females: 5-12 mcg/dL or 71-142 nmol/L
- Total T3: 80-180 ng/dL

Vitamins

Normal vitamin ranges include the following [8] :

- Folate (serum): 5-25 ng/mL or 11-57 nmol/L
- Vitamin A: 32.5-78.0 µg/dL
- Vitamin B12: 200-800 pg/mL
- Vitamin C: 0.4-1.5 mg/dL
- Vitamin D: 1,25-Dihydroxyvitamin D: 15-60 pg/mL; 25-Hydroxyvitamin D: 30-60 ng/mL [9]
- Vitamin E
- Serum: 5-20 mcg/mL (11.6-46.4 µmol/L)
- Plasma: 5.7-19.9 mg/L (11.4-39.8 µmol/L)
- Vitamin K: 0.2-1.0 ng/ml

Tumor markers

Normal tumor marker ranges include the following [8] :

- α fetoprotein: Adults: < 40 ng/mL or < 40 mcg/L; β human chorionic gonadotropin: females, premenopausal nonpregnant: < 1.0 U/L, postmenopausal: < 7.0 U/L; males: < 1.4 U/L
- CA19.9: 0-37 U/mL
- Carcinoembryonic antigen: < 2.5 ng/mL
- Prostatic acid phosphatase: 0-3 U/dL
- Prostate-specific antigen: 0-2.5 ng/mL

Miscellaneous

Additional normal ranges include the following [8, 9] :

- α 1-antitrypsin: 85-213 mg/dL or 0.85-2.13 g/L
- Angiotensin-converting enzyme: 8-53 U/L
- C-reactive protein: 1.0 mg/dL or < 10 mg/L
- D-dimer: 500 µg/L or < 0.5 mg/L Fibrinogen Equivalent Units
- Erythrocyte sedimentation rate: Males: < 15 mm/h; females: ≤ 20 mm/h
- Lactate dehydrogenase: 100-190 units/L at 37° C (lactate - pyruvate) or 100-190 units/L
- Lead: < 10 µg/dL
- Rheumatoid factor: Negative (< 60 units/mL by nephelometric testing)

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